

Robust Personalized Movie Recommendation System Based on Embedding Technology

Xinyi Zhou

Faculty of Science and Technology
Beijing Normal-Hong Kong Baptist University
Zhuhai, Guangdong, China
2386194226@qq.com

Abstract

Personalized recommendation is essential for navigating the vast landscape of online content. Here we present an embedding-based Neural Collaborative Filtering framework for movie recommendation, designed to address the scalability and personalization limitations of traditional methods. Using the MovieLens dataset, we implement a negative sampling strategy to construct implicit feedback signals, and jointly learn low-dimensional user and item embeddings through a multi-layer perceptron to capture high-order, non-linear interactions. The model achieves a Hit Ratio at rank 10 of 0.86, substantially improving top-N recommendation accuracy. This work demonstrates the effectiveness of embedding-driven deep learning for large-scale, personalized movie recommendation and offers a modular foundation for future integration with transfer learning, sequential modeling, and large language models.

CCS Concepts

• **General and reference** → Document types; General conference proceedings.

Keywords

movie recommendation system, word embedding, neural collaborative filtering, deep learning

ACM Reference Format:

Xinyi Zhou. 2025. Robust Personalized Movie Recommendation System Based on Embedding Technology. In *2025 3rd International Conference on Internet of Things and Cloud Computing Technology (IoTCCCT 2025)*, September 27–29, 2025, Haikou, China. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3776865.3776922>

1 Introduction

In the era of big data, with the rapid increase in information volume, personalized services have become crucial for meeting user needs. Recommendation systems, as one of the key technologies in this field, have been widely applied across various industries, especially in e-commerce, social platforms, news aggregation, and streaming services [1]. By analyzing users' historical behaviors and preferences, recommendation systems provide personalized content recommendations, significantly enhancing user experience while also driving commercial conversion for platforms. They help users

discover content that might interest them and, through precise recommendations, increase user engagement and drive platform value. Therefore, recommendation systems have become a core technology for improving user satisfaction and enhancing platform competitiveness.

As a form of digital entertainment content, movies have gradually become one of the most important content categories on online platforms with the development of internet technologies. Compared to other types of content, movies offer rich genres, complex plots, and diverse audience demands. As a result, accurately recommending movies that match user interests has become a crucial aspect of the development of movie recommendation systems (MRS) [2]. An effective movie recommendation system not only helps users easily discover movies they enjoy but also pushes personalized movie content based on users' historical behavior and viewing preferences, thereby enhancing the overall viewing experience.

However, traditional movie recommendation methods largely rely on similarity-based calculations between users and movies, such as collaborative filtering and content-based recommendation techniques [3]. While these methods can provide initial recommendations in certain scenarios, their limitations become evident when dealing with large-scale user data and vast movie resources. For instance, collaborative filtering-based methods perform poorly when handling cold-start problems, such as when there is insufficient historical data on users or movies. On the other hand, content-based methods are often constrained by movie tags and descriptions, making it difficult to capture users' deeper preferences. Furthermore, traditional methods tend to overlook the complex semantic relationships between movies, leading to a lack of diversity and personalization in recommendation results. These issues present significant challenges in the practical application of traditional recommendation methods.

To address these challenges, recommendation systems based on deep learning have garnered increasing attention in recent years. Deep learning techniques, particularly neural network-based models, are capable of extracting complex nonlinear relationships between users and movies from large datasets, thus improving the accuracy and effectiveness of recommendation systems. Embedding techniques, a crucial method in deep learning, map the features of movies and users into a low-dimensional vector space, where similar users and movies are clustered together. This approach not only improves computational efficiency but also significantly enhances the capability of recommendation systems to handle high-dimensional data and complex relationships. By leveraging embedding techniques, movie recommendation systems can better



This work is licensed under a Creative Commons Attribution 4.0 International License. *IoTCCCT 2025, Haikou, China*

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-2114-4/2025/09

<https://doi.org/10.1145/3776865.3776922>

capture users' deeper interests, offering more personalized recommendations.

In this paper, we propose a movie recommendation system based on embedding technology. The system explores the deep semantic relationships between movies and users to provide personalized recommendations for each user, effectively addressing the limitations of traditional recommendation methods. Our system is not only capable of efficiently operating in large-scale data environments but also offers higher recommendation accuracy and stronger personalization, significantly improving the user experience. Experimental validation demonstrates that the embedding-based movie recommendation system proposed in this study excels in recommendation accuracy, providing new insights and methods for further development in the field of movie recommendation systems.

2 Previous Work

Currently, there are several popular types of movie recommendation systems, mainly including users-based collaborative filtering, item-based collaborative filtering, content-based recommendation systems and hybrid filtering systems. These traditional approaches remain foundational in many platforms due to their interpretability and relatively low computational complexity. However, they are lack of personalization and have other insurmountable limitations. User-based collaborative filtering (UBCF) recommends items to a target user by identifying other users with similar rating behaviors [4]. While it is intuitive and effective in small to medium sized datasets, UBCF suffers from scalability issues and poor performance in sparse environments, particularly when user-item interactions are limited. Item-based collaborative filtering (IBCF) identifies items similar to those a user has previously liked, based on collective user ratings [5]. This method is more stable over time and typically more scalable than UBCF. However, it still encounters challenges in cold-start scenarios, especially for newly added items with insufficient interaction data. Content-based recommendation systems analyze item features, such as genre, director, cast, or textual metadata, and match them to user profiles to generate suggestions. This approach performs well in situations involving new or niche items and avoids dependence on the behavior of other users. Nonetheless, it may lead to over-specialization, where users are repeatedly exposed to similar content, limiting diversity and novelty. Hybrid recommendation systems combine collaborative and content-based techniques, aiming to overcome the individual drawbacks and leverage the advantages of both. They tend to offer more accurate and robust recommendations, particularly in addressing cold-start and sparsity problems. However, the integration of multiple algorithms introduces additional complexity and often requires fine-tuning and larger computational resources.

With the rise of deep learning, more advanced and expressive recommendation models have emerged. Convolutional Neural Networks (CNNs) have been successfully applied to extract visual or semantic features from auxiliary content such as posters, trailers, or plot descriptions, thereby enriching content-based models with high-level representations. However, CNNs are less suitable for capturing user behavior over time. To address sequential dependencies, Recurrent Neural Networks (RNNs) have been employed to model dynamic user preferences based on temporal interactions.

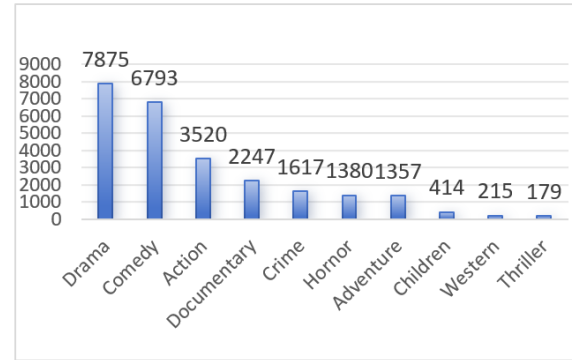


Figure 1: Quantity of genres in MovieLens dataset

These models are particularly effective in session-based or time-aware recommendation tasks, though they often require substantial training time and are sensitive to sequence length. Moreover, deep hybrid models such as Neural Matrix Factorization [6] and Deep Factorization Machine [7] represent a new generation of recommendation algorithms that integrate deep learning architectures with collaborative signals. These models are capable of capturing both low- and high-order feature interactions, offering superior predictive performance. However, they typically demand large-scale datasets, careful hyperparameter tuning, and considerable computational resources. Therefore, this paper focus on analyzing the movie recommendation system that based on word embedding technology that realize personalization to the greatest extent.

3 Dataset and Preprocessing

The data used for analysis in this paper is collected from MovieLens dataset [8]. The MovieLens dataset, curated and maintained by the GroupLens Research Lab at the University of Minnesota, is one of the most widely used benchmark datasets in the field of recommendation systems research. It provides a rich set of user-movie interactions, enabling robust experimentation and evaluation of collaborative filtering, content-based, and hybrid recommendation algorithms. Multiple versions of the dataset are available, varying in size to suit different experimental needs. Each dataset typically includes users, which are represented by anonymous userIDs; movie metadata such as titles, genres, release years, and movie IDs; and the rating history such as timestamps and rating values assigned by users to movies. The dataset is pre-processed and anonymized, ensuring user privacy while maintaining sufficient complexity for algorithmic development. Researchers frequently use MovieLens to validate models under controlled experimental setups and to test cold-start, sparsity, and scalability scenarios. In this work, we choose a rating range from 0 to 5 and adopt "1" as the threshold, that is, considering ratings greater than "1" as positive interactions; otherwise, they are negative instances. The numbers and rating distributions of different types of movies are shown in Figures 1 and 2

To prepare the data for model training, we performed a structured preprocessing pipeline on the MovieLens dataset. Specifically,

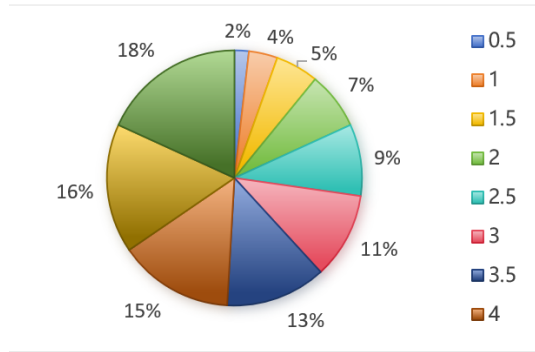


Figure 2: The distribution of rating score in MovieLens dataset

we randomly sampled approximately 30% of users to reduce computational overhead while maintaining user diversity. For each selected user, their interaction history was temporally ordered, and the most recent interaction information was reserved for testing, with the remaining interactions forming the training set. Adopting an implicit feedback paradigm, we binarized all training ratings by assigning a value of 1 to indicate observed user-item interactions, which are regarded as positive instances. To construct negative instances for training, we employed negative sampling: for each positive user-item pair, we randomly selected four items that the user had not interacted with, ensuring they were not in the part of the user’s historical viewing records, and labeled these as 0. This process resulted in a balanced binary classification dataset comprising positive and sampled negative examples, serving as supervised signals for training personalized recommendation models.

4 Model and Results

The proposed model adopts a Neural Collaborative Filtering paradigm, implemented within the PyTorch Lightning framework (Figure 3 and Table 1), to predict the probability of user-item interactions by capturing and modelling the intricate, non-linear relationships between users and items. At its core, the architecture begins with two learnable embedding layers: one dedicated to user representations and the other to movie representations. These layers transform high-cardinality, discrete identifiers into continuous-valued latent vectors, thereby enabling the model to operate within a compact and semantically rich representation space. Firstly, we use one-hot encoding method to encode users and movies. The outcomes are then inputted into a neural network for embedding method. Specifically, the user embedding layer maps each unique user identifier to an 8-dimensional latent vector, while the item embedding layer performs an analogous mapping for each item identifier. The embeddings are initialized stochastically and are jointly optimized with the remainder of the network via backpropagation, such that they capture latent user and item characteristics that extend beyond directly observable features.

For each user-item pair under consideration, the corresponding embeddings are retrieved and concatenated to yield a unified 16-dimensional representation vector. This joint vector serves as the

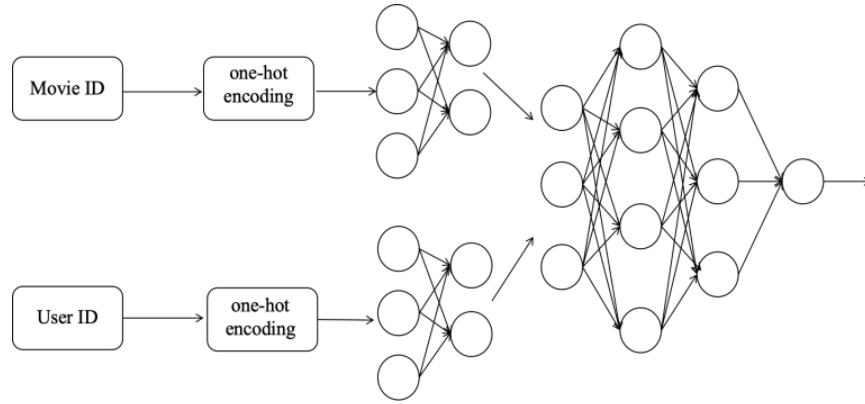
input to a multi-layer perceptron designed to model complex, high-order interactions between latent factors. The first fully connected (FC) layer projects the 16-dimensional input into a 64-dimensional feature space, followed by a Rectified Linear Unit (ReLU) activation to introduce non-linearity and enable the capture of intricate behavioral patterns not accessible to purely linear models. The subsequent dense layer reduces this representation to 32 dimensions, again employing a ReLU activation to preserve non-linear modelling capacity. The final transformation is a single-unit dense layer, producing a scalar output which is passed through a sigmoid activation function. This final non-linearity maps the scalar to the interval $[0, 1]$, yielding an interpretable probability that reflects the predicted likelihood of a positive interaction between the specified user and item.

Model training is driven by the minimization of the binary cross-entropy loss function, which quantifies the divergence between predicted interaction probabilities and ground-truth binary labels, with “1” denoting an observed (positive) interaction and “0” indicating the absence thereof. Parameter updates across the embedding layers and dense layers are performed via Adaptive Moment Estimation (Adam) optimization algorithm, which adaptively tunes individual learning rates for each parameter, thereby facilitating both efficient convergence and robustness to noisy gradients. This end-to-end learning process ensures that the model not only captures salient latent factors but also learns to combine them in a manner that maximally discriminates between positive and negative user-item interactions in the training data.

To evaluate the performance of the recommendation model, we employ the Hit Ratio at rank 10 (Hit Ratio@10) as the primary metric. For each user-item pair in the test set, the model constructs a recommendation list by randomly sampling 99 items that the user has not previously interacted with, then adding the one actual test item, which the user interacted with most recently to this list, resulting in a candidate set of 100 items. The model predicts the preference scores for these 100 items through output probability outcomes and ranks them in descending order. If the true test item appears in the top 10 of the ranked lists, it can be considered as a “hit”. The final Hit Ratio at rank 10 is calculated as the proportion of test set for which the correct item was successfully retrieved within the top 10 recommendations. In our experiment, this method yields a Hit Ratio at rank 10 of 0.86, indicating that the model is able to include the correct item in the top 10 recommendations for 86% of the users.

5 Discussion

This study introduces a personalized recommendation system based on Neural Collaborative Filtering trained on the MovieLens dataset. By embedding users and items in a shared latent space and learning non-linear interaction functions via a multi-layer perceptron, the model effectively captures complex preference structures that traditional linear or factorization-only approaches may miss. Under a leave-one-out evaluation protocol, the system attains a Hit Ratio at rank 10 of 0.86, indicating strong performance in top-N recommendation. The implementation in PyTorch Lightning emphasizes reproducibility and modularity, offering a practical foundation for further research and production deployment.

**Figure 3: Model Framework****Table 1: Model Framework**

Layer	Name	Description	Params
1	user embedding	Embedding	1M
2	item embedding	Embedding	1M
3	fc1	Linear	1K
4	Relu1	Relu	0
5	fc2	Linear	2K
6	Relu2	Relu	0
7	output	Linear	33

Despite these strengths, several limitations merit attention. First, the dataset offers limited contextual signals about users (e.g., age, occupation) and items (e.g., rich synopses), constraining the model’s ability to leverage content features or situational context for recommendation. Second, the genre distribution is highly imbalanced, with popular categories far more prevalent than niche ones; this induces a long-tail bias whereby the model preferentially learns dominant genres while under-serving underrepresented categories. Third, although the NCF architecture is computationally tractable, training time grows with model depth, embedding dimensionality, and the negative sampling strategy; scaling to larger corpora and near-real-time applications will require additional engineering and algorithmic advances.

Future research can address these limitations through richer modeling and data augmentation. A promising direction is to integrate transfer learning and self-supervised pretraining. Large-scale contrastive objectives, such as maximizing agreement between co-occurring user-item pairs versus randomly sampled pairs, can shape user and item embeddings before supervised fine-tuning. Masked-interaction or masked-sequence pretraining, where items in a user’s history are masked and predicted, can endow the model with stronger sequential priors. Cross-domain transfer is also attractive: embeddings pretrained on broader consumption logs (e.g., multi-platform media interactions) can be adapted to MovieLens via domain-adversarial alignment or parameter-efficient fine-tuning, mitigating cold-start and long-tail sparsity.

Large language models (LLMs) [9], reinforcement learning (RL) [10], temporal and session-based architectures [11], graph neural networks (GNNs) [12], and bias-aware training [13] jointly offer a rich design space for advancing recommender systems. On the item side, LLMs can transform unstructured text, such as plot summaries, reviews, synopses, or external metadata, into dense descriptors aligned with collaborative embeddings; on the user side, they can summarize long-form feedback, infer stable preferences and transient intents, and produce natural-language rationales that foster user trust. LLM-derived signals can be injected as features to two-tower retrieval models, prompts to rerankers that refine NCF scores with text-aware judgments, or lightweight cross-encoders for shortlists, with parameter-efficient adapters or low-rank updates reducing tuning cost. RL provides a principled framework for optimizing long-term value rather than short-term proxies, employing contextual bandits for safe exploration, off-policy learning with conservative Q-learning, slate-aware list modeling, and doubly robust evaluation to bridge offline–online gaps; integrating NCF embeddings as state features improves sample efficiency, while coupling RL with LLM-generated explanations allows shaping reward functions for diversity, coverage, and serendipity. Sequential models, particularly Transformer-based architectures (SASRec- or BERT-style), capture order, recency, and periodicity, with hybrid designs combining sequential encoders and NCF heads to model both long-term taste and short-term intent; continual-learning strategies (regularization, replay, parameter isolation) mitigate catastrophic forgetting under distribution shift, and distilled

or linear-time variants meet strict latency budgets. GNNs exploit higher-order relations in user–item graphs via LightGCN propagation, knowledge-graph enrichment, hypergraph co-consumption modeling, and attention-based neighbor selection, with meta-paths encoding domain knowledge and fusion with LLM text embeddings producing robust multi-modal representations. Long-tail and genre imbalance challenges are addressed through re-weighted or focal losses, curriculum learning for rare classes, hard-negative/adversarial sampling, and exposure-aware objectives with counterfactual risk minimization to debias from historical policies. Cold-start and few-shot settings benefit from MAML-style meta-learning, zero-shot retrieval from text-aligned embeddings, and multi-modal contrastive learning from visual or audio artifacts. Evaluation should extend beyond Hit Ratio @10 to ranking fidelity (NDCG, MAP), calibration, novelty, coverage, and diversity, with robustness tests on subpopulation shifts, tail-only slices, and sequence truncation, and with careful off-policy evaluation and online experiments for methods affecting exposure. From a systems perspective, scalable deployments combine ANN-based retrieval with expressive rerankers (e.g., cross-attention or LLM-augmented), apply mixed-precision training, quantization-aware inference, and teacher-student distillation, and maintain freshness via streaming updates; privacy-preserving learning through federated optimization with secure aggregation and differential privacy enables edge personalization under data constraints.

6 Conclusion

In this work, we developed a personalized movie recommendation system based on Neural Collaborative Filtering with embedding technology, aiming to overcome the limitations of traditional similarity-based approaches. Leveraging the MovieLens dataset, we designed a structured preprocessing pipeline, including negative sampling, to construct a balanced training set for implicit feedback modeling. Our architecture jointly learns user and item embeddings and captures high-order, non-linear interactions through a multi-layer perceptron, enabling richer representation of user preferences. Experimental results demonstrate that the proposed model achieves a Hit Ratio@10 of 0.86 under a leave-one-out evaluation protocol, indicating strong top-N recommendation performance.

This study not only validates the effectiveness of embedding-based deep learning methods for movie recommendation but also provides a modular, reproducible framework that can be extended with advanced techniques, such as transfer learning, large language model integration, and sequential modeling, to further enhance personalization, scalability, and robustness in real-world applications.

References

- [1] H. Ko, S. Lee, Y. Park, and A. Choi, "A survey of recommendation systems: Recommendation models, techniques, and application fields," *Electronics*, vol. 11, no. 1, p. 141, Jan. 2022, doi: 10.3390/electronics11010141.
- [2] M. Goyani and N. Chaurasiya, "A review of movie recommendation system: Limitations, survey and challenges," *ELCVIA Electron. Lett. Comput. Vis. Image Anal.*, vol. 19, pp. 18–37, 2020, doi: 10.5565/rev/elcvia.1232.
- [3] H. Papadakis, A. Papagrigoriou, C. Panagiotakis, E. Kosmas, and P. Fragopoulou, "Collaborative filtering recommender systems taxonomy," *Knowl. Inf. Syst.*, vol. 64, no. 1, pp. 35–74, Jan. 2022, doi: 10.1007/s10115-021-01628-7.
- [4] H. Wang, Z. Shen, S. Jiang, G. Sun, and R.-J. Zhang, "User-based collaborative filtering algorithm design and implementation," *J. Phys. Conf. Ser.*, vol. 1757, no. 1, p. 012168, Jan. 2021, doi: 10.1088/1742-6596/1757/1/012168.
- [5] C. Ajaegbu, "An optimized item-based collaborative filtering algorithm," *J. Ambient Intell. Humaniz. Comput.*, vol. 12, no. 12, pp. 10629–10636, Dec. 2021, doi: 10.1007/s12652-020-02876-1.
- [6] Liu, H., Wang, W., Zhang, Y., Gu, R., & Hao, Y. (2022). Neural matrix factorization recommendation for user preference prediction based on explicit and implicit feedback. *Computational Intelligence and Neuroscience*, 2022(1), 9593957.
- [7] Q. Wang, M. Zhang, Y. Zhang, J. Zhong, and V. S. Sheng, "Location-based deep factorization machine model for service recommendation," *Appl. Intell.*, vol. 52, no. 9, pp. 9899–9918, July 2022, doi: 10.1007/s10489-021-02998-9.
- [8] H. Maxwell and K. A., "The MovieLens datasets," *ACM Trans. Interact. Intell. Syst. TiiS*, Dec. 2015, doi: 10.1145/2827872.
- [9] K. Bao, J. Zhang, Y. Zhang, W. Wenjie, F. Feng, and X. He, "Large language models for recommendation: Progresses and future directions," in *Proceedings of the Annual International ACM SIGIR Conference on Research and Development in Information Retrieval in the Asia Pacific Region*, Beijing China: ACM, Nov. 2023, pp. 306–309. doi: 10.1145/3624918.3629550.
- [10] M. M. Afsar, T. Crump, and B. Far, "Reinforcement learning based recommender systems: A survey," *ACM Comput. Surv.*, vol. 55, no. 7, pp. 1–38, July 2023, doi: 10.1145/3543846.
- [11] R. Wang, J. Lou, and Y. Jiang, "Session-based recommendation with time-aware neural attention network," *Expert Syst. Appl.*, vol. 210, p. 118395, Dec. 2022, doi: 10.1016/j.eswa.2022.118395.
- [12] C. Gao, X. Wang, X. He, and Y. Li, "Graph neural networks for recommender system," in *Proceedings of the Fifteenth ACM International Conference on Web Search and Data Mining*, Virtual Event AZ USA: ACM, Feb. 2022, pp. 1623–1625. doi: 10.1145/3488560.3501396.
- [13] L. Boratto and M. Marras, "Advances in bias-aware recommendation on the web," in *Proceedings of the 14th ACM International Conference on Web Search and Data Mining*, Virtual Event Israel: ACM, Mar. 2021, pp. 1147–1149. doi: 10.1145/3437963.3441665.